

Exam Questions: "Attention Is All You Need"

Section 1: Conceptual Understanding (20 questions)

Basic Concepts

- 1. What is the main innovation introduced by the Transformer architecture?**
 - Expected answer: Replacing recurrent and convolutional layers entirely with self-attention mechanisms
- 2. Explain why traditional RNN models face challenges with parallelization during training.**
 - Expected answer: RNNs process sequences sequentially, where each hidden state depends on the previous one, preventing parallel computation within training examples
- 3. What are the two main components of the Transformer architecture?**
 - Expected answer: Encoder and Decoder, each consisting of stacked layers with self-attention and feed-forward networks
- 4. Define self-attention (intra-attention) and explain its purpose.**
 - Expected answer: Self-attention is an attention mechanism that relates different positions of a single sequence to compute a representation, allowing the model to weigh the importance of different words when processing each position
- 5. What problem does positional encoding solve in the Transformer?**
 - Expected answer: Since Transformers have no recurrence or convolution, positional encoding injects information about the relative or absolute position of tokens in the sequence

Architecture Details

- 6. Describe the structure of a single encoder layer in the Transformer.**
 - Expected answer: Multi-head self-attention mechanism followed by a position-wise feed-forward network, with residual connections and layer normalization around each sub-layer
- 7. How does the decoder differ from the encoder in the Transformer?**
 - Expected answer: The decoder has three sub-layers (instead of two): masked self-attention, encoder-decoder attention, and feed-forward network. The masking prevents positions from attending to subsequent positions
- 8. Why is masking necessary in the decoder's self-attention layer?**
 - Expected answer: To preserve the auto-regressive property and prevent leftward information flow, ensuring predictions for position i depend only on known outputs at positions less than i
- 9. What is the purpose of residual connections in the Transformer?**
 - Expected answer: To facilitate gradient flow during training and enable the training of deeper networks
- 10. Explain the role of layer normalization in the Transformer architecture.**
 - Expected answer: Normalizes the inputs across features to stabilize training and improve convergence

Attention Mechanism

- 11. What are Query (Q), Key (K), and Value (V) in the attention mechanism?**
 - Expected answer: Q represents "what we're looking for," K represents "what information each position contains," and V represents "the actual information to retrieve." They're all learned linear projections of the input
- 12. Write and explain the formula for Scaled Dot-Product Attention.**
 - Expected answer: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$
 - Explanation: Compute dot products of query with all keys, scale by $\sqrt{d_k}$, apply softmax to get weights, then multiply by values
- 13. Why is the dot product scaled by $1/\sqrt{d_k}$ in the attention formula?**
 - Expected answer: For large d_k values, dot products grow large in magnitude, pushing softmax into regions with extremely small gradients. Scaling counteracts this effect
- 14. What is Multi-Head Attention and why is it beneficial?**
 - Expected answer: Instead of single attention, linearly project Q, K, V multiple times (h times) with different learned projections, perform attention in parallel, then concatenate. Allows the model to jointly attend to information from different representation subspaces at different positions

15. **How many attention heads does the base Transformer model use, and what are the dimensions?**

- Expected answer: $h = 8$ heads, with $d_k = d_v = d_{\text{model}}/h = 64$ for each head

Training and Optimization

16. **What optimizer was used to train the Transformer, and what were the key hyperparameters?**

- Expected answer: Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$, and $\epsilon = 10^{-9}$

17. **Describe the learning rate schedule used in the Transformer.**

- Expected answer: Learning rate increases linearly for the first warmup_steps (4000) training steps, then decreases proportionally to the inverse square root of the step number

18. **What regularization techniques were employed during training?**

- Expected answer: Residual dropout ($P_{\text{drop}} = 0.1$), dropout on embeddings and positional encodings, and label smoothing ($\epsilon_{\text{ls}} = 0.1$)

19. **What is label smoothing and why was it used?**

- Expected answer: Smoothing target distributions ($\epsilon_{\text{ls}} = 0.1$) makes the model more uncertain, hurting perplexity but improving accuracy and BLEU score, preventing overconfident predictions

20. **How were sentence pairs batched during training?**

- Expected answer: Batched by approximate sequence length, with each batch containing approximately 25,000 source tokens and 25,000 target tokens

Section 2: Technical Analysis (15 questions)

Complexity and Efficiency

21. **Compare the computational complexity per layer of self-attention vs. recurrent layers.**

- Expected answer: Self-attention: $O(n^2 \cdot d)$, Recurrent: $O(n \cdot d^2)$. Self-attention is faster when sequence length $n <$ representation dimension d

22. **What is the maximum path length between any two positions in the input/output for self-attention layers?**

- Expected answer: $O(1)$ - constant, which helps with learning long-range dependencies

23. **Explain why the Transformer is more parallelizable than RNN-based models.**

- Expected answer: Self-attention connects all positions with operations that can be computed simultaneously, whereas RNNs require $O(n)$ sequential operations

24. **What is the trade-off of using restricted self-attention for very long sequences?**

- Expected answer: Reduces complexity from $O(n^2 \cdot d)$ to $O(r \cdot n \cdot d)$ but increases maximum path length from $O(1)$ to $O(n/r)$

25. **How does the computational cost of multi-head attention compare to single-head attention?**

- Expected answer: Similar total cost because the dimension of each head is reduced ($d_k = d_{\text{model}}/h$), compensating for the multiple heads

Position-wise Feed-Forward Networks

26. **What is the structure of the position-wise feed-forward network?**

- Expected answer: Two linear transformations with ReLU activation: $\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$

27. **What are the dimensionalities in the feed-forward network?**

- Expected answer: Input/output dimension $d_{\text{model}} = 512$, inner layer dimension $d_{\text{ff}} = 2048$

28. **Why is the feed-forward network called "position-wise"?**

- Expected answer: Applied to each position separately and identically, using the same parameters across positions but different parameters across layers

Positional Encoding

29. **Write the formulas for positional encoding used in the Transformer.**

- Expected answer:
 - $\text{PE}(\text{pos}, 2i) = \sin(\text{pos}/10000^{(2i/d_{\text{model}})})$
 - $\text{PE}(\text{pos}, 2i+1) = \cos(\text{pos}/10000^{(2i/d_{\text{model}})})$

30. **Why did the authors choose sinusoidal positional encoding over learned embeddings?**

- Expected answer: May allow the model to extrapolate to sequence lengths longer than those encountered during training, and they produced nearly identical results

31. How are positional encodings incorporated into the model?

- Expected answer: Added to the input embeddings at the bottoms of the encoder and decoder stacks

Model Variants

32. What is the difference between the base model and the big model?

- Expected answer: Big model has $d_{\text{model}} = 1024$ (vs. 512), $d_{\text{ff}} = 4096$ (vs. 2048), $h = 16$ heads (vs. 8), and uses dropout rate 0.3 (vs. 0.1)

33. According to Table 3, what happens when you use too many or too few attention heads?

- Expected answer: Single-head attention is 0.9 BLEU worse than optimal ($h=8$), but quality also drops with too many heads ($h=16$ or $h=32$)

34. What did experiments with learned positional embeddings reveal?

- Expected answer: Nearly identical results to sinusoidal positional encoding (row E in Table 3)

35. How did reducing the attention key size d_k affect model quality?

- Expected answer: Hurt model quality, suggesting that determining compatibility is not easy and a more sophisticated function than dot product may be beneficial

Section 3: Results and Applications (10 questions)

Machine Translation Results

36. What BLEU score did the big Transformer achieve on WMT 2014 English-to-German translation?

- Expected answer: 28.4 BLEU, improving over previous best by more than 2.0 BLEU

37. How long did it take to train the big Transformer model?

- Expected answer: 3.5 days on 8 P100 GPUs (300,000 steps)

38. What was significant about the training cost of the Transformer compared to previous models?

- Expected answer: Achieved superior results at a fraction of the training cost (e.g., base model used 3.3×10^{18} FLOPs vs. competitors using 10^{19} - 10^{20} FLOPs)

39. What beam search parameters were used during inference?

- Expected answer: Beam size of 4, length penalty $\alpha = 0.6$, maximum output length = input length + 50

40. How did the authors obtain final models for evaluation?

- Expected answer: For base models, averaged last 5 checkpoints (10-minute intervals); for big models, averaged last 20 checkpoints

English Constituency Parsing

41. Why was English constituency parsing a good test for the Transformer's generalization ability?

- Expected answer: Presents specific challenges - strong structural constraints on output, output significantly longer than input, and RNN models hadn't achieved state-of-the-art in small-data regimes

42. What F1 score did the 4-layer Transformer achieve on WSJ parsing in the semi-supervised setting?

- Expected answer: 92.7 F1 on Section 23

43. How did the Transformer compare to the BerkleyParser on WSJ-only training?

- Expected answer: Outperformed BerkleyParser even when training only on the WSJ training set of 40K sentences

Dataset Information

44. What datasets were used for training the translation models?

- Expected answer: WMT 2014 English-German (4.5M sentence pairs with 37K token vocabulary) and English-French (36M sentences with 32K token vocabulary)

45. What encoding method was used for the training data?

- Expected answer: Byte-pair encoding for both language pairs

Section 4: Comparative Analysis (10 questions)

- 46. **Compare the advantages of self-attention over recurrent layers.**
 - Expected answer: Lower sequential operations ($O(1)$ vs $O(n)$), constant maximum path length, better parallelization, and faster when $n < d$
- 47. **What are the three desiderata for comparing different layer types mentioned in Section 4?**
 - Expected answer: (1) Total computational complexity per layer, (2) Amount of parallelizable computation, (3) Path length between long-range dependencies
- 48. **How does the Transformer handle long-range dependencies compared to RNNs?**
 - Expected answer: Constant $O(1)$ path length vs $O(n)$ for RNNs, making it easier to learn long-range dependencies
- 49. **What advantage does the Transformer have over ConvS2S?**
 - Expected answer: Constant operations to relate any two positions vs linear growth with distance; also achieved better BLEU scores with lower training cost
- 50. **Why might self-attention yield more interpretable models?**
 - Expected answer: Individual attention heads learn to perform different tasks and exhibit behavior related to syntactic and semantic structure (shown in visualizations)

Section 5: Application Questions (10 questions)

Three Applications of Attention

- 51. **Describe how attention is used in the encoder-decoder attention layers.**
 - Expected answer: Queries come from previous decoder layer, keys and values come from encoder output, allowing decoder to attend over all input positions
- 52. **Explain self-attention layers in the encoder.**
 - Expected answer: Keys, values, and queries all come from the same place (output of previous encoder layer), allowing each position to attend to all positions in the previous layer
- 53. **How do self-attention layers work in the decoder differently from the encoder?**
 - Expected answer: Each position can only attend to positions up to and including itself (not future positions), implemented by masking out illegal connections (setting them to $-\infty$)

Architecture Choices

- 54. **Why do the embedding layers multiply weights by $\sqrt{d_{\text{model}}}$?**
 - Expected answer: To maintain appropriate scale for the embeddings before adding positional encodings
- 55. **What weight sharing is done in the model?**
 - Expected answer: The same weight matrix is shared between the two embedding layers (encoder and decoder) and the pre-softmax linear transformation

Historical Context

- 56. **What were the dominant sequence transduction models before the Transformer?**
 - Expected answer: Recurrent neural networks (LSTM and GRU) and convolutional neural networks with encoder-decoder architectures
- 57. **What previous work attempted to reduce sequential computation?**
 - Expected answer: Extended Neural GPU, ByteNet, and ConvS2S, all using CNNs as building blocks
- 58. **How did the Transformer differ from previous attention mechanisms?**
 - Expected answer: First transduction model relying entirely on self-attention without using sequence-aligned RNNs or convolution
- 59. **What challenge did recurrent models face that motivated the Transformer?**
 - Expected answer: Sequential nature precluded parallelization within training examples, critical at longer sequence lengths due to memory constraints
- 60. **What was the significance of the attention visualizations in the appendix?**
 - Expected answer: Showed that attention heads learn interpretable patterns like anaphora resolution, syntactic relationships, and long-distance dependencies

Section 6: Critical Thinking (15 questions)

- 61. **Discuss the trade-off between model size and performance based on Table 3.**
 - Expected answer: Bigger models perform better (row C), but there's a point of diminishing returns with attention heads; optimal configuration balances performance and computational cost
- 62. **Why might additive attention outperform dot-product attention for large d_k values without scaling?**
 - Expected answer: Large dot products push softmax into regions with small gradients; additive attention with a feed-forward network doesn't have this issue
- 63. **What would be the limitations of using the Transformer for very long sequences?**
 - Expected answer: $O(n^2 \cdot d)$ complexity makes it computationally expensive; restricted self-attention could help but increases path length
- 64. **How does the auto-regressive property of the decoder affect inference speed?**
 - Expected answer: Decoder must generate one token at a time, consuming previously generated tokens as input, limiting parallelization during inference
- 65. **Why is dropout particularly important for the Transformer?**
 - Expected answer: Table 3 shows dropout is "very helpful in avoiding over-fitting" given the model's capacity
- 66. **Explain how Multi-Head Attention provides benefits over single-head attention.**
 - Expected answer: Multiple heads allow jointly attending to different representation subspaces and learning diverse relationships; single attention head averaging inhibits this
- 67. **What might be challenges in applying the Transformer to modalities beyond text?**
 - Expected answer: Need to handle continuous data, positional information might work differently, computational cost for high-dimensional data like images
- 68. **Why did the authors plan to explore restricted attention mechanisms?**
 - Expected answer: To efficiently handle large inputs/outputs like images, audio, and video where full $O(n^2)$ attention would be prohibitively expensive
- 69. **Discuss the significance of the Transformer achieving state-of-the-art with "significantly less time to train."**
 - Expected answer: Makes advanced models more accessible, enables faster research iteration, reduces computational costs and environmental impact
- 70. **How might the constant path length in self-attention help with gradient flow?**
 - Expected answer: Shorter paths reduce vanishing/exploding gradient problems, making it easier to learn long-range dependencies
- 71. **Why might the model share weights between embeddings and pre-softmax transformation?**
 - Expected answer: Reduces parameters, provides consistency between input/output representations, may help with learning
- 72. **What insights do the attention visualizations provide about model behavior?**
 - Expected answer: Show interpretable patterns (anaphora resolution, syntactic structure, long-distance dependencies), demonstrating what the model learns beyond just performance metrics
- 73. **Evaluate the choice of using 8 attention heads. What if they used 4 or 16?**
 - Expected answer: Based on Table 3, 8 heads (BLEU 25.8) outperforms both fewer and more heads; 4 heads (25.5) slightly worse, 16 heads (25.1) notably worse
- 74. **Why might the Transformer generalize well to English constituency parsing despite no task-specific tuning?**
 - Expected answer: Self-attention naturally captures syntactic and semantic structure; architecture is flexible enough to learn different sequence-to-sequence mappings
- 75. **What are the implications of the Transformer being "the first transduction model relying entirely on self-attention"?**
 - Expected answer: Paradigm shift in NLP; showed recurrence isn't necessary for sequence modeling; influenced subsequent models like BERT, GPT, and modern LLMs

Section 7: Short Answer & Essay Questions (10 questions)

- 76. **Explain the complete forward pass of an input through the Transformer encoder.**
 - Expected: Token embeddings → add positional encoding → N layers of (multi-head self-attention → add & norm → feed-forward → add & norm) → encoder output
- 77. **Describe the complete process of generating a translated sentence using the Transformer.**
 - Expected: Encode source sentence → initialize decoder with start token → generate tokens auto-regressively (decoder attends to encoder output and previous tokens) → stop at end token
- 78. **Compare and contrast the computational efficiency of self-attention, recurrent, and convolutional layers.**

- Expected: Complexity comparison (Table 1), parallelization capabilities, path lengths, and practical implications for different sequence lengths
 - 79. Discuss the importance of the scaling factor in Scaled Dot-Product Attention with mathematical reasoning.**
 - Expected: Explain variance of dot products grows with d_k , impact on softmax gradients, mathematical derivation in footnote, and practical consequences
 - 80. Analyze the results in Table 2 and explain what makes the Transformer superior to previous models.**
 - Expected: Higher BLEU scores, lower training costs (FLOPs), specific comparisons to models like GNMT, ConvS2S, MoE
 - 81. Explain how residual connections and layer normalization contribute to training deep Transformers.**
 - Expected: Residual connections enable gradient flow, prevent vanishing gradients; layer normalization stabilizes training, both essential for 6+ layer depth
 - 82. Describe the three different ways multi-head attention is used in the Transformer.**
 - Expected: (1) Encoder self-attention, (2) Decoder masked self-attention, (3) Encoder-decoder attention - explain each with purpose and mechanism
 - 83. Evaluate the Transformer's performance on both machine translation and parsing tasks. What does this demonstrate?**
 - Expected: State-of-the-art on translation, strong on parsing despite no tuning; demonstrates generalization ability and versatility of attention-based architecture
 - 84. Discuss potential future directions for the Transformer architecture based on the conclusion.**
 - Expected: Other modalities (images, audio, video), local/restricted attention for efficiency, less sequential generation, applications beyond NLP
 - 85. Critically analyze the design choices made in the Transformer architecture. What are potential weaknesses or areas for improvement?**
 - Expected: $O(n^2)$ complexity limitation, auto-regressive decoder slow at inference, fixed positional encoding limitations, potential improvements from later research
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Answer Key Guidelines

For Grading:

- Conceptual questions (1-20): 2-3 marks each
- Technical questions (21-45): 3-4 marks each
- Application questions (46-60): 4-5 marks each
- Critical thinking (61-75): 5-7 marks each
- Essay questions (76-85): 10-15 marks each

Evaluation Criteria:

- Accuracy of technical details
- Clarity of explanation
- Understanding of underlying concepts
- Ability to connect different parts of the paper
- Critical analysis and insight